

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1436

Perryman 138 kV
November 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1436. The Interconnection Customer proposes a 28.7 MW solar photovoltaic facility interconnecting at the Perryman 138 kV substation in the Maryland area. The request was received on March 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 424 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$61,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Center-E Towns 138.0 kV	Coldspring-Recovery Soco 138.0 kV line	111.3	400	5.0
Coldspring-Recovery Soco 138.0 kV Line	Ctr-East Towns 138.0 kV Line	105.3	403	5.4
Ctr-Coldspring 138.0 kV	E Towson-Ctr 138.0 kV L	104.2	347	5.4
Ctr-Highgrove 138.0 kV	Sierra Pacific Lumber-Riva Road Substation 345.0 kV	101.6	519	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Towson-Center 138.0 kV	Rebuild line with higher-capacity conductor	\$4,033,000	1% (\$39,000)
Coldspring Substation-Recovery Soco Substation 138.0 kV L	Reconductor line	\$1,350,000	1% (\$12,000)
Ctr-Coldspring 138.0 kV	Reconductor line	\$387,000	1% (\$4,000)
Highgrove-Center 138.0 kV	Reconductor line	\$705,000	1% (\$6,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$61,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.